TEMASEK JUNIOR COLLEGE PRELIMINARY EXAMINATION
H2 GEOGRAPHY (PAPER 2) SUGGESTED ANSWER SCHEME

1 (a) Possible responses include: [4]

Strength:

- Clearly defined relationship
- Feasible and manageable
- Linked to geographical theory
- Researchable and measurable

Weaknesses:

- The hypothesis does not specify *how flood risk is defined* (e.g. frequency, severity, or impact of floods). This may create ambiguity in measurement.
- It assumes a *linear downstream increase*, when flood risk may actually fluctuate depending on local channel characteristics, land use, or floodplain settlement (e.g. contrast between Sites 5 and 6).

(b) Possible responses include: [6]

At **Site 1 (Resource 2A)**:

- *Risk*: Uneven, stony riverbed may cause slips or falls while measuring depth.
  Minimisation: Students should wear appropriate footwear with good grip and work in pairs to stabilise each other.

At **Site 5 (Resource 2B)**:

- *Risk*: Dense vegetation and steep banks create trip hazards and limited access.
  Minimisation: Use designated access points, and brief students on safe movement around vegetation.
- *Risk*: Insects or exposure to allergens near vegetation.
  Minimisation: Students should wear long sleeves and insect repellent.

(c) Possible responses include:

At **Site 1 (Resource 2A)**, the stream is narrow and shallow, with a visibly stony, steep-sided channel. These upper-course characteristics limit the volume of water that can accumulate, so large-scale flooding is less likely. However, the steep valley sides encourage rapid surface runoff, so the main flood risk here would be short-lived flash floods confined to the channel. [4]

At **Site 5 (Resource 2B)**, the stream flows through a wider, flatter valley floor with dense vegetation on the banks. Although the channel is slightly wider and deeper, the gentler gradient means water can spread laterally onto the floodplain during heavy rainfall. The presence of vegetation may slow flow, but it also indicates proximity to human access areas (e.g., footpaths or settlements). This makes Site 5 more prone to damaging overbank flooding that can affect surrounding land uses.

(d) Possible responses include:

Strengths: [6]

- The questionnaire includes closed, pre-coded responses makes the data easy to quantify and compare between respondents.
- Q1 and Q2 collect background information allows analysis of how perceptions vary spatially and by exposure to the river.

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- Q5 and Q6 assess awareness and trust in flood management, providing insight into the social dimension of flood risk.

Limitations:

- The Likert scale compatibility to conventional coding. Respondents may misinterpret the scale, leading to inaccurate or unreliable data.
- Questions are largely perception-based, which may not reflect actual hydrological flood risk.
- Conducting the survey on a weekday afternoon in September limits representativeness, as only certain demographics are more likely to be available.

(e) **Possible responses include:** [5]

A **line graph** or a **bar graph** can be used to represent the changes in wetted perimeter downstream along the Hawkcombe stream.

- **X-axis**: Site number (1–6, representing upstream to downstream).
- **Y-axis**: Wetted perimeter (metres).
- **Title**: "Changes in Wetted Perimeter Downstream along the Hawkcombe Stream".

(f) **Possible responses include:**

- **Discharge measurements** (velocity × cross-sectional area) for more accurate flood potential. [5]
- **Rainfall data** from gauges to link precipitation to discharge.
- **Soil infiltration rates** to assess runoff potential.
- **Land use mapping** (e.g., urbanisation, impermeable surfaces) to evaluate human influence.
- **Bankfull capacity** or frequency of past floods recorded from secondary sources.

Possible responses include:

(g) The students' investigation into flood risk along the Hawkcombe stream provides valuable insights into how channel characteristics and perceptions vary downstream, but the validity of their findings is constrained by methodological weaknesses. [10]

Strengths lie in its clear geographical framework. The flood risk map (Resource 1) anchors the study in real spatial variation, showing how risk zones change along the catchment. Physical data on channel width and depth at six sites (Resource 4) allow calculation of wetted perimeter, directly linking to hydrological theory such as Bradshaw's Model, which predicts increasing discharge downstream. Photographs of Site 1 and Site 5 (Resources 2A and 2B) illustrate how physical setting can be related to overbank flooding. The questionnaire (Resource 3) adds a human dimension, exploring residents' perceptions of likelihood, concern, awareness, and trust. The use of closed questions and Likert scales produces quantifiable data that can be compared across locations, while combining physical and perceptual evidence strengthens the geographical scope of the investigation.

However, several limitations reduce the reliability and accuracy of conclusions. Only six sites were sampled, which may not capture the variability of tributary inputs, land use, or human modification along the stream. Channel measurements were taken using a single depth reading and an unweighted tape laid across the bed, a method prone to error in uneven channels and thus

undermining data accuracy. The resident survey introduces additional issues: it was carried out on a weekday afternoon in September, which likely excludes working adults and younger groups, creating demographic bias. Moreover, the inverted Likert scales risk confusing respondents, which can distort results when coding or analysing the data. Beyond methodological flaws, the scope of the investigation is narrow: it equates flood risk mainly with channel size and resident perception, overlooking key influences such as rainfall intensity, infiltration capacity, land use change, and historical flood frequency.

Overall, while the investigation demonstrates sound geographical design by integrating maps, field measurements, and surveys, its conclusions—particularly the claim that Site 5 faces the highest flood risk—should be treated cautiously. The evidence suggests that flood risk does increase downstream, but methodological weaknesses in sampling, reliability, and accuracy mean that a more comprehensive dataset, including discharge measurements, rainfall records, and broader survey coverage, would be necessary to make a definitive judgment.

2 (a) Possible responses include: [4]

In 1990, only a few cities across Africa had populations above 1 million, mostly concentrated in the north (e.g. Cairo) and along coastal areas of West and Southern Africa.

By 2018, the number of such cities increased significantly, with a wider spread across the continent. There was notable growth in West Africa (e.g. Lagos), East Africa (e.g. Nairobi), and parts of Southern Africa, showing a shift from limited concentration to a more dispersed continental distribution.

(b) Possible responses include: [5]

- **Africa**: Highest reliance on open dumps (~60%), limited use of sanitary landfills or recycling.
- **Asia & Pacific**: Mix of sanitary landfills (~40%) and open dumps (~30%), with some incineration and recycling.
- **Europe & Central Asia**: Most waste sent to sanitary landfills and recycling/incineration, very little to open dumps.
- **Latin America & Caribbean**: Sanitary landfills dominate, though open dumps still account for ~25%.
- **North America**: Heavily dependent on sanitary landfills, with incineration and recycling making up a significant share.
  → Overall, developed regions (Europe, North America) have more advanced waste management (sanitary, recycling), while developing regions (Africa, parts of Asia) still rely on open dumping.

(c) Possible responses include: [5]

Recycling reduces the need for raw material extraction, lowering environmental degradation. For instance, recycling aluminium saves up to 95% of the energy required compared to mining bauxite, reducing deforestation and soil erosion in areas such as Guinea. Paper recycling decreases demand for logging, protecting forests in Brazil and Canada. Recycling plastics reduces oil demand, though effectiveness depends on collection systems. However, recycling itself has limits: energy use in processing, transport emissions, and low recycling rates in some regions mean that pressure on natural environments is reduced but not eliminated.

(d) Possible responses include:

The cartoon shows how poorer countries face disproportionate impacts (e.g. drought, flooding, food insecurity), despite contributing least to emissions. Low-income countries lack financial resources, technology, and infrastructure to adapt. For example, Bangladesh faces annual flood losses of ~2% of GDP, while wealthier countries like the Netherlands invest billions in flood defences such as dikes and storm surge barriers. High-income countries can also diversify economies and insure against losses, reducing vulnerability. Thus, the same climate hazard has more severe social and economic impacts in low-income countries. [5]

(e) Possible responses include: [5]

Urban reimaging revitalises cities to attract investment, tourism, and talent. For example, Bilbao's Guggenheim Museum transformed the city into a cultural hub, boosting tourism revenue and jobs. In Singapore, Marina Bay Sands and Gardens by the Bay repositioned the city as a global financial and tourism hub, supporting economic diversification. In London, the Docklands redevelopment attracted financial services, replacing industrial decline. Economic reasons include maintaining global competitiveness, regenerating post-industrial areas, and enhancing quality of life to retain skilled workers.

(f) Possible responses include: [6]

Bosco Verticale, or the "Vertical Forest," is a pioneering residential complex located in Milan, Italy. Designed by the Italian architect Stefano Boeri and completed in 2014, it represents a groundbreaking approach to integrating nature within urban living spaces. The project challenges traditional concepts of architecture and urban design by incorporating thousands of trees and plants directly into the façade of high-rise buildings. Bosco Verticale stands as a bold statement about the possibilities of sustainable urban development, biodiversity enhancement, and climate adaptation in densely populated cities. Bosco Verticale in Milan represents a transformative approach to urban living that integrates architecture and ecology in unprecedented ways.

- **Mitigation**: Trees absorb CO_2, reducing emissions; vegetation filters particulate matter, improving air quality; shading reduces building energy demand for cooling.
- **Adaptation**: Vertical greenery reduces the urban heat island effect, provides insulation against extreme temperatures, and creates habitats for biodiversity within the city. The design also enhances residents' well-being and resilience to climate extremes. Thus, it demonstrates how built environment innovations can serve dual roles in tackling climate change.

While challenges remain in maintenance and scalability, the project stands as a beacon for future cities seeking harmony between built environments and nature. As urban populations continue to grow globally, Bosco Verticale offers an inspiring model for creating resilient, healthy, and green cities of the future.

QUESTION 3

Possible Approaches:

Liveability has become an increasingly important concept in contemporary society due to rapid urbanization, environmental challenges, social inequalities, and the growing recognition that quality of life is central to sustainable development. At its core, liveability refers to the overall quality of an area or city as a place to live, encompassing factors such as access to basic services, safety, environmental quality, economic opportunities, social cohesion, and cultural amenities. The importance of liveability today arises from the way it directly impacts human well-being, health,

economic productivity, and social stability, as well as broader environmental sustainability. This essay explores the reasons why liveability has gained prominence in the modern world, emphasizing demographic trends, urban challenges, health considerations, environmental concerns, and socio-economic dimensions. Candidates could approach the question by evaluating the relative importance of social issues compared to other challenges in affecting urban liveability. They could explore specific social issues such as inequality, crime, housing shortages, access to services, and social exclusion, and assess how these impact residents' quality of life in urban areas. Candidates could consider case studies from different cities or urban regions to illustrate how social challenges manifest in diverse contexts. They might evaluate the success or failure of urban policies aimed at addressing social issues, such as affordable housing schemes, crime reduction strategies, or community development programs. Candidates could also analyse the interplay between social issues and other factors — for example, how environmental degradation or inadequate infrastructure exacerbate social problems, or how economic challenges influence social cohesion and liveability. Candidates might also assess the role of governance, planning, and community participation in managing social challenges and improving liveability, and consider the difficulties in resolving deep-rooted social inequalities.

Marking:

- Responses should demonstrate clear understanding of the concept of urban liveability and the various factors influencing it.
- High-level answers will use relevant, well-developed case studies to support evaluation.
- Candidates should engage critically with the statement, showing balance and nuance in their argument.
- Analysis of the complexity and interrelationships between social and other urban challenges is expected.
- Evaluation should include a reasoned judgment supported by evidence and examples.
- Use Generic Level Descriptors for 20-mark H2 essays.

QUESTION 4

Possible Approaches:

Tackling climate change has become one of the most urgent and critical challenges facing humanity today. The importance of addressing climate change lies not only in its far-reaching environmental impacts but also in its profound effects on human health, economic stability, social equity, and global security. Climate change is driven primarily by the accumulation of greenhouse gases (GHGs) such as carbon dioxide, methane, and nitrous oxide in the atmosphere, largely resulting from human activities like fossil fuel combustion, deforestation, and industrial processes. These emissions intensify the natural greenhouse effect, leading to rising global temperatures, altered weather patterns, and disruptions across ecological and human systems. Candidates could approach the question by identifying and evaluating different key actors involved in climate change responses, such as national governments, international organisations, non-governmental organisations (NGOs), multinational corporations, local governments, and civil society groups. They could analyse the roles each actor plays in mitigation and adaptation strategies. Candidates could evaluate the effectiveness of these actors by considering criteria such as policy implementation, influence on global agreements, funding and resources mobilised, innovation in technology, and ability to engage stakeholders. They could also assess the extent to which each actor has achieved tangible outcomes in reducing greenhouse gas emissions or enhancing resilience to climate impacts. Candidates might explore case studies illustrating successes and limitations of various actors. For example, the effectiveness of the Paris Agreement (international governments), corporate commitments to carbon neutrality, grassroots activism driving policy change, or local government initiatives in climate adaptation. Candidates could analyse challenges faced by these actors, such as political will, economic constraints, conflicting interests, enforcement issues, and public engagement. Candidates could also assess how collaboration between actors influences overall

effectiveness, and whether some actors have greater leverage in shaping global or local climate outcomes.

Marking:

- Responses should show clear understanding of the diverse actors involved in climate change responses and their roles.
- High-level answers will support evaluation with detailed, relevant examples and case studies.
- Candidates should critically analyse the relative strengths and weaknesses of different actors.
- Evaluation should consider both mitigation and adaptation efforts.
- The answer should demonstrate awareness of challenges and constraints faced by key actors.
- A reasoned, balanced judgment on the relative effectiveness of actors is expected.
- Use Generic Level Descriptors for 20-mark H2 essays.

Reminder of essay level descriptors

Levels	Marks	Generic Level Descriptors for H2 Essays
5	18–20	Evaluation is consistently analytical and coherent. Response is well-supported by relevant material, including the effective use of examples. Response features accurate geographical knowledge and reflects good understanding of the subject content relevant to the question.
4	14–17	Evaluation is analytical and coherent. Response is mostly well-supported by relevant material, including the appropriate use of examples. Response features accurate geographical knowledge and reflects adequate understanding of the subject content relevant to the question.
3	10–13	Evaluation is broadly analytical and generally coherent. Response is moderately well-supported by relevant material, including some appropriate use of examples. Response features accurate geographical knowledge and reflects adequate understanding of the subject content relevant to the question.
2	6–9	Response is largely descriptive with limited analysis and evaluation. Response is partly coherent and may lack clarity in parts. Response is poorly supported by relevant materials, including the limited use of examples. Response features inaccurate geographical knowledge and poor understanding of the subject content relevant to the question.
1	1–5	Response is descriptive with no analysis or evaluation. Response is fragmented and lacks clarity. Response consists of unsupported assertions. Response features largely inaccurate geographical knowledge and a lack of understanding of the subject content relevant to the question.
0	0	No creditworthy response